

Rate–Capacity Threshold: Instability Conditions in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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Abstract

This document defines instability conditions as a function of load accumulation rate relative to capacity in Dimensional Human Field Theory (DHFT).

Instability occurs when load accumulates at a rate that exceeds capacity under boundary conditions permitting continued accumulation.

I. Scope

This document establishes the relation between the rate of load accumulation rate and capacity.

It defines threshold conditions under which instability occurs without requiring instantaneous load to exceed capacity.

II. Definition

Rate–Capacity Threshold is the condition under which the rate of load accumulation exceeds capacity.

The rate of load accumulation is expressed as dL/dt , representing change in load over time.

III. Threshold Condition

Instability occurs when:

$$dL/dt > C$$

under boundary conditions permitting continued accumulation.

IV. Non-Equivalence of Rate and Magnitude

Instability does not require total load to exceed capacity at a single measurement point.

Load remains below capacity while accumulating load at a rate that produces instability over time.

V. Temporal Consequence

Rate-based instability produces:

- delayed instability
- divergence between measured state and structural condition
- accumulated load sufficient for later constraint violation

VI. Constraint

All rate-based conditions reduce to relations between:

- Load (L)
- Capacity (C)
- Boundary (B)

No additional variables are introduced.

VII. Statement

Instability occurs when load accumulation rate exceeding capacity constraints, independent of instantaneous load magnitude.

VIII. System Reference

The system defined in this document is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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